R22

17/11/2025

Code No.: 22CSC15N

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.E. (CSE, CET) V Sem (Main/Backlog) Examination November 2025

Operating Systems

Time: 3 Hours **Max Marks: 60**

Note: "Answer **ALL** questions from **Part-A** and **Part-B** (including **internal choices**) at one place and in the same order as in the question paper."

Part - A

(5Q X 2M = 10 Marks)

			M	CO	BT
1		Compare Desktop systems and mobile systems.	(2)	1	2
2		What are the uses of job queues, ready queues and device queues?	(2)	2	1
3		Consider a logical address space of 64 pages of 1,024 words each, mapped onto a physical memory of 32 frames. How many bits are there in the logical address, and physical address?	(2)	3	1
4		Is it possible to have a deadlock involving only one single-threaded process? State the reason.	(2)	4	1
5		What are the advantages and disadvantages of providing mandatory locks instead of advisory locks whose use is left to users' discretion?	(2)	5	1

Part - B

(5Q X 10M = 50 Marks)

			M	CO	BT
6	(a)	Explain different operations performed by the operating system.	(5)	1	2
	(b)	Describe some of the challenges of designing operating systems for mobile devices compared with designing operating systems for traditional PCs.	(5)	1	3
		(OR)			
7	(a)	Explain about the dual mode operation in OS with a neat block diagram.	(5)	1	2
	(b)	Discuss, with examples, how the problem of maintaining coherence of cached data manifests itself in the following processing environments: Distributed systems.	(5)	1	2
8	(a)	What is PCB? Describe its framework with clear applications of each field in process execution.	(5)	2	2
	(b)	Some computer systems provide multiple register sets. Describe what happens when a context switch occurs if the new context is already loaded into one of the register sets. What happens if the new context is in memory rather than in a register set and all the register sets are in use?	(5)	2	3
		(OR)			
9	(a)	Describe the state transition mechanisms of process.	(5)	2	2
	(b)	Suppose that the following processes arrive for execution at the times indicated. Each process will run for the amount of time listed. In answering the questions, use nonpreemptive scheduling, and base all decisions on the information you have at the time the decision must be made.	(5)	2	3

Process	Arrival Time	Burst Time
P_1	0.0	8
P_2	0.4	4
P_3	1.0	1

a. What is the average turnaround time for these processes with the SJF scheduling algorithm?

The SJF algorithm is supposed to improve performance, but notice that we chose to run process P1 at time 0 because we did not know that two shorter processes would arrive soon. Compute what the average turnaround time will be if the CPU is left idle for the first 1 unit and then SJF scheduling is used. Remember that processes P1 and P2 are waiting during this idle time, so their waiting time may increase. This algorithm could be called future-knowledge scheduling.

10	(a)	On a system with paging, a process cannot access memory that it does not own. Why? How could the operating system allow access to additional memory? Why should it or should it not?	(5)	3	2
	(b)	Explain why address-space identifiers (ASIDs) are used in TLBs.	(5)	3	2
		(OR)			
11	(a)	What is the effect of allowing two entries in a page table to point to the same page frame in memory? Explain how this effect could be used to decrease the amount of time needed to copy a large amount of memory from one place to another. What effect would updating some byte on one page have on the other page?	(5)	3	3
	(b)	Consider the following page reference string: 7, 2, 3, 1, 2, 5, 3, 4, 6, 7, 7, 1, 0, 5, 4, 6, 2, 3, 0 , 1. Assuming demand paging with three frames, how many page faults would occur for the Optimal replacement algorithms?	(5)	3	3
12	(a)	Explain why spinlocks are not appropriate for single-processor systems yet are often used in multiprocessor systems.	(5)	4	2
	(b)	Show that, if the wait() and signal() semaphore operations are not executed atomically, then mutual exclusion may be violated.	(5)	4	2
		(OR)			
13	(a)	Describe how deadlock is possible with the Bounded-Buffer problem.	(5)	4	2
	(b)	Assume that a multithreaded application uses only reader–writer locks for synchronization. Applying the four necessary conditions for deadlock,is deadlock still possible if multiple reader–writer locks are used?	(5)	4	3
14	(a)	Describe the various file organization methods with illustrative examples.	(5)	5	2
	(b)	Explain file directory structure, and caching optimizations.	(5)	5	2
		(OR)			
15	(a)	Buffer-overflow attacks can be avoided by adopting a better programming methodology or by using special hardware support. Discuss these solutions.	(5)	5	3
	(b)	Discuss the strengths and weaknesses of implementing an access matrix using capabilities that are associated with domains.	(5)	5	2

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